$$F_{Ey\max} = 0.385k\frac{qQ}{d^2} \ . \qquad (4)$$

Charged particle P feels both an upward F_{Ey} and a downward gravitational force mg. Only if mg equals the Coulomb y component is P in equilibrium. If P has weight mg greater than $F_{Ey\max}$, P cannot be in equilibrium. For mass m an equilibrium requires

$$mg \leqslant F_{Ey\max} \ .$$

From (4) we obtain

$$m \leqslant \frac{0.385}{g}\ k\frac{qQ}{d^2} \ . \qquad (5)$$

2. $y\,(m) > 0.707d$; $0 < y\,(m) \leqslant 0.707d$.

3. By the given condition, P is in stable equilibrium at $y = y_1$; hence

$$k\frac{Qqy_1}{(d^2 + y_1^2)^{\frac{3}{2}}} - m_1 g = 0 \ . \qquad (6)$$

Give P a small y-displacement $\triangle y$, so P feels a net y-directed force

$$F_y = F_{Ey} - m_1 g = k\frac{Qq\,(y_1+\triangle y)}{[\,d^2+(y_1+\triangle y)^2\,]^{3/2}} - m_1 g \ . \qquad (7)$$

Since $\triangle y$ is small, neglecting higher-order terms gives

$$F_y = k\frac{Qq\,(y_1+\triangle y)}{[\,d^2+y_1^2+2y_1\triangle y\,]^{3/2}} - m_1 g$$

$$= k\frac{Qq\,(y_1+\triangle y)}{(d^2+y_1^2)^{3/2}}\left(1 - \frac{3y_1\triangle y}{d^2+y_1^2}\right) - m_1 g$$

$$= k\frac{Qqy_1}{(d^2+y_1^2)^{3/2}} + k\frac{Qq\triangle y}{(d^2+y_1^2)^{3/2}} - k\frac{3qQy_1^2\triangle y}{(d^2+y_1^2)^{5/2}} - m_1 g \ .$$

Using (6) gives

$$F_y = -\frac{m_1 g\,(2y_1^2 - d^2)}{(d^2+y_1^2)\,y_1}\triangle y \ . \qquad (8)$$

Since $y = y_1$ is stable for P, $y_1 > 0.707d$, $2y_1^2 - d^2 > 0$. From (8) implies

P along y has restoring force: small oscillations about equilibrium are harmonic. Their angular frequency is

$$\omega = \sqrt{\frac{(2y_1^2 - d^2)}{(d^2+y_1^2)\,y_1}g}\ , \qquad (9)$$